

Infrastructure as Code with Copilot

DevOps & Infrastructure | Advanced

Agenda

Time	Topic
0:00–0:08	IaC model
0:08–0:20	Terraform module
0:20–0:30	Bicep comparison
0:30–0:42	Module composition
0:42–0:52	Security review
0:52–1:00	Validation and lab handoff

IaC model

Concept	Terraform	Bicep
Resource	<code>resource</code> block	<code>resource</code> declaration
Reusable unit	<code>module</code>	<code>module</code>
Input	<code>variable</code>	<code>param</code>
Output	<code>output</code>	<code>output</code>
Preview	<code>terraform plan</code>	<code>az deployment what-if</code>

IaC describes desired resources, not intent

The same configuration can create a secure private service or an exposed one. The difference is in the details: identity, network paths, defaults, and dependencies.

Before generating code, establish:

- the architecture boundary and approved regions;
- the inputs that come from the environment;
- who owns state and applies changes;
- the controls that must be visible in review.

Copilot can draft resources. The architecture must come first.

Modules create contracts

A module is useful when callers should depend on a stable interface rather than the resources inside it.

Module element	Review question
Input	Is this the smallest safe configuration surface?
Output	Does this expose only what callers need?
Resource	Does it meet the architecture and policy constraints?
Documentation	Can another team use it without reading its internals?

Avoid modules that only hide a single resource without creating a useful boundary.

State and preview answer different questions

State records what the IaC tool believes it manages. A plan or what-if compares the requested configuration with the target environment.

Both need protection:

State: approved backend, access controls, no secrets in source

Preview: approved identity, bounded environment, human review before apply

Never use generated IaC as proof that a deployment is safe. Use the preview to find changes, then review whether those changes should happen.

Prompt from facts

Read this Terraform project. Add a network module for the supplied CIDRs.
Expose only needed VPC and subnet IDs. Use variables and outputs; do not add
credentials, public database access, or a plan/apply command.

Check every resource against the architecture before accepting it.

Security review

- Block unneeded public access.
- Restrict ingress by port and source.
- Keep secrets out of source.
- Encrypt storage and databases.
- Limit IAM actions and resources.
- Add required logging.

Generated IaC is a draft. Human review owns the deployment decision.

Validate locally

```
terraform fmt  
terraform validate  
az bicep build --file main.bicep
```

A plan or what-if needs an approved environment, credentials, and stop guard.

Lab

Generate Terraform and Bicep, secure the supplied Terraform, and compare approved Space context with a normal prompt. Do not provision cloud resources.